Where a y^+-only closure for P/ε stops working, and what a stress-based one fixes

Sidharth SS
Independent Researcher

21 October 2026

Abstract

Most local algebraic closures for the production-to-dissipation ratio P/ε are written as functions of the wall distance y^+ alone. Two of them, M16 and PySR, are accurate to about one percent over the near-wall region, and on the standard inner-layer test mask they are indistinguishable from each other and from a seven-constant alternative. We score four closures on 52 direct numerical simulation and large-eddy simulation profiles from nine public databases, using the full profile rather than the inner layer. On that test M16 and PySR both return negative R^2 (-0.160 and -0.167), while a four-constant closure built around the total stress, SURGA, returns 0.640. The reason is structural: a function of y^+ has no way of knowing how far the wall is from the edge of the layer, so it is forced to a constant value in the outer region, and its error grows with the Reynolds number. We then test the same closures on two classes they were not built for, a compressible channel up to Mach 2 and an adverse-pressure-gradient boundary layer, and report where every one of them, including SURGA, still fails.

1 The problem

Reynolds-averaged models need the ratio of turbulent kinetic energy production to its dissipation, P/ε. In the logarithmic layer this ratio is very close to one, which is why it is often taken as one. Away from the log layer it is not close to one: it peaks near 1.8 in the buffer region, dips below one, and then either decays towards zero at the edge of the flow or rises again in the outer layer, depending on the flow. A constant value is wrong over most of the profile.

A local algebraic closure tries to do better without adding a transport equation. The input is whatever is known at a point. Two of the expressions in current use are

$$\text{M16:}\quad \frac{P}{\varepsilon} = \frac{\tanh^2(0.111\,y^+)}{\tanh(0.052\,y^+ - 0.443) + 2.888/y^+}, \tag{1}$$

$$\text{PySR:}\quad \frac{P}{\varepsilon} = \frac{\tanh(0.128\,y^+)}{\tanh(0.053\,y^+ - 0.620) + 5.78/y^+}, \tag{2}$$

each with four fitted constants and each a pure function of y^+. Both were obtained by fitting high-quality channel and boundary-layer data, both are cheap, and both reproduce the buffer-layer peak closely. The question this paper asks is a simple one: how far do they hold once you leave the region they were fitted on?

2 Data and how we score

We use 52 profiles taken from nine public databases (Table 1). The only inclusion rule was that the dataset must publish a dissipation budget, because P/ε cannot be formed from mean and Reynolds-stress data alone. No profile was reconstructed by fitting a dissipation model, and no value was interpolated across datasets. Each ratio was checked against its own kinetic-energy budget before use.

Table 1: The nine databases. Every profile comes from the primary provider.

Flow class	Source	Profiles
Channel	Bernardini, Pirozzoli & Orlandi (2014)	5
Channel	Lee & Moser (2015)	5
Channel	Moser, Kim & Mansour (1999)	3
Couette	Lee & Moser (2018)	3
Pipe	El Khoury et al. (2013)	4
ZPG boundary layer	Schlatter & Örlü (2010)	9
Compressible channel	Gerolymos & Vallet (2014)	9
APG boundary layer	Bobke et al. (2017)	10
Wing (APG)	NACA4412 LES (KTH)	4
Total		**52**

Three test masks are used throughout. The *inner mask*, $1 < y^+ \leq \min(150, 0.3\,Re_\tau)$, is the one on which these closures are usually reported. The *full mask*, $y^+ \geq 1$ with $\eta = y/\delta \leq 1$, covers the whole profile from the buffer layer to the edge. A third mask starting at $y^+ = 0.05$ is used as a harness check. Goodness of fit is R^2 of P/ε, averaged over profiles within a flow class.

3 The closures in current use

On the inner mask all four closures land within a single percentage point of each other: 0.876 for M16, 0.867 for PySR, 0.875 for a seven-constant alternative, SEMBAN, and 0.873 for SURGA. This is the number that gets quoted, and it is a fair number: inside $y^+ < 150$ the four expressions really are interchangeable, because they were all calibrated to reproduce the same buffer-layer peak. Their peaks are 1.808, 1.832, 1.760 and 1.771 at $y^+ = 11.4$, 11.9, 11.8 and 12.2 respectively, and the measured peak in the datasets is between 1.7 and 1.85.

The conclusion usually drawn from numbers like these is that the problem is solved. It is not. The inner mask covers at most 150 wall units out of a layer that can be several thousand, and it truncates exactly the part of the profile where the four expressions differ. A closure can be perfect on this mask and useless in a RANS calculation, which is evaluated over the whole field.

4 Where the y^+-only closures break

Table 2 repeats the comparison on the full profile.

Table 2: Full mask, $y^+ \geq 1$ and $\eta \leq 1$. Mean R^2 per flow class. Best per row in bold.

Flow class	n	M16	PySR	SEMBAN	SURGA
All	52	−0.160	−0.167	−0.474	0.640
Channel	13	0.251	0.264	−0.162	0.972
Pipe	4	0.621	0.639	0.426	0.954
Couette	3	0.887	0.897	0.775	0.653
ZPG boundary layer	9	−2.346	−2.337	−3.612	0.963
APG boundary layer	10	0.174	0.114	0.286	0.077
Compressible channel	9	0.423	0.433	0.383	0.777
Wing (APG)	4	−0.282	−0.344	−0.097	−0.393

Two things change at once. The overall scores collapse, and they collapse unevenly. M16

and PySR go from 0.876 and 0.867 on the inner mask to −0.160 and −0.167 on the full profile. On the nine zero-pressure-gradient boundary layers they reach −2.35 and −2.34, which is worse than predicting the mean. SURGA, which has fewer constants than SEMBAN, holds 0.640 overall.

The failure is not a matter of coefficient choice. Every expression of the form $f(y^+)$ tends to a constant as $y^+ \to \infty$; M16 and PySR both tend to about 0.98–1.00. But the measured P/ε in the outer layer of a channel falls to zero, in a boundary layer it falls to about 0.1, and in an adverse-pressure-gradient boundary layer it does not fall at all (Section 6.2). The closure cannot tell these cases apart, because they have the same y^+ and different η. No choice of the four constants can fix that: the missing argument is the one that does not appear in the formula.

The practical consequence is scale-dependent. A y^+-only law is accurate at low Reynolds number, where the inner layer is most of the profile, and increasingly wrong at high Reynolds number, where the outer region dominates. Figure 1 shows this on the 13 incompressible channels. SURGA is flat in Re_τ; the other three decay monotonically and cross zero.

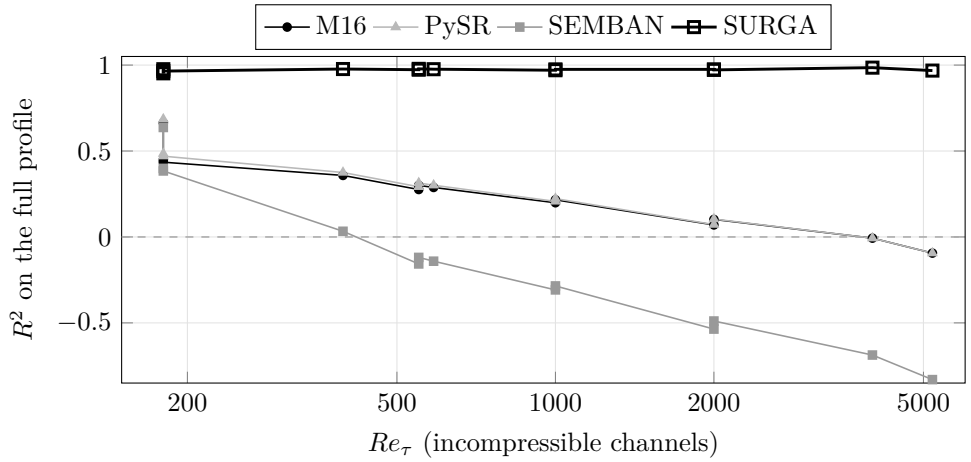


Figure 1: R^2 against friction Reynolds number for 13 channel profiles. The y^+-only closures lose accuracy as the outer layer grows; the stress-based closure does not.

5 What the new closure adds

SURGA keeps four fitted constants, fewer than SEMBAN, and is written as a product of four factors:

$$\frac{P}{\varepsilon} = \underbrace{\tanh^3(a_1^e y^+)}_{\text{gate}} \underbrace{\left[1 - T^5(1-T^2) + c_1 y^{+2} e^{-b_1 y^+}\right]}_{\text{inner}} \times \underbrace{\left[\frac{\tanh(A(\tau+t_e))}{\tanh(A(1+t_e))}\right]^2}_{\text{window}} \underbrace{\exp\left[-c_c \frac{Re_\tau}{1000}\,\eta^4 \min(\tau,1)^2\right]}_{\text{core}} \tag{3}$$

with $T = \tanh(b_1^2 y^+/2)$, $a_1^e = a_1(1+c_v V_w)$, $t_e = \min(\tau|_{\eta=1}, 1)$, and $b_1 = 1.1a_1$, $c_1 = 0.4a_1$ tied to a_1 rather than fitted separately. The fitted constants are $a_1 = 0.19113$, $A = 5.076$, $c_c = 0.87330$ and $c_v = 2.32391$.

The gate reproduces the cubic approach to the wall. The inner factor carries the buffer peak and, importantly, a trough that is not fitted: the function $1 - T^5(1-T^2)$ has a minimum at $T^* = \sqrt{5/7} = 0.8452$ with depth $1 - 50\sqrt{35}/2401 = 0.8768$, so the dip after the peak sits at a fixed place and depth determined by the algebra rather than by the data.

The last two factors are the ones that matter here. The *window* follows the total stress τ: production is proportional to the stress still available at that height, and the ratio should follow it. The *core* factor uses Re_τ and η, so the closure knows both how deep in wall units the point is and how far it is from the edge. That is the information the y^+-only laws lack, and it is why SURGA can fall to zero at a channel centreline while staying finite near the wall of a boundary layer. None of this is extra data: τ, η and Re_τ are all available in any RANS field.

6 Results

Table 2 is the main result. SURGA is the only closure with a positive R^2 over all 52 profiles, and it is best in four of the seven flow classes. The gains are largest where the old closures fail worst: channels (0.972 against 0.250 for M16), pipes (0.954 against 0.621) and zero-pressure-gradient boundary layers (0.963 against -2.346). It is worth noting that this is not a consequence of more constants: SEMBAN has seven and returns -0.474 overall.

The closure was not built for either of the two classes we added in this study.

6.1 Compressible channel

Nine channel cases at centreline Mach numbers from 0.35 to 2.0 and Re_τ^* from 65 to 341 give SURGA 0.777 on the full mask, against 0.423 and 0.433 for the two y^+-only closures. Nothing was added to the closure for compressibility.

The one systematic change with Mach number is where the peak sits: $y^+ \approx 12$ at $M = 0.35$ and $y^+ \approx 28$ at $M = 2.0$, while the peak height barely moves (1.86 to 1.96). A closure with the peak pinned at $y^+ = 11.4$ cannot follow that, which is most of the reason the two y^+-only laws drop to about 0.43 here. SURGA follows it only partly: the peak in its inner factor is also fixed, and the shift is picked up by the window and core factors rather than by the peak term itself.

6.2 Adverse pressure gradient

This is the harder test, and it needed two geometries to interpret. We have ten flat-plate large-eddy simulation stations covering five pressure-gradient histories and $\beta = 0.88$ to 4.40, plus four stations on a NACA4412 wing where β reaches 4.56. Fourteen profiles in total.

The first thing to report is what the near-wall peak does. At $\beta = 0.96$ the measured peak is 1.73 and M16 gives 1.81. At $\beta = 4.40$ the measured peak is 1.64, below the zero-pressure-gradient value. So the peak, on a flat plate, is not suppressed and not amplified by the pressure gradient, and the models handle it correctly. The peaks of 2.15 to 2.67 seen on the wing are therefore not a pressure-gradient effect at all: they come from wall curvature and from a stress history that has not reached equilibrium. A wing-only dataset cannot separate those two causes; a β-matched flat plate can.

The second thing is where the profile actually departs. Table 3 averages P/ε over $0.2 \leq \eta \leq 0.8$, the outer layer.

For a channel the outer layer sits at 0.87 and for a pipe at 0.81. For a zero-pressure-gradient boundary layer it is 1.10. For an adverse-pressure-gradient boundary layer it is 1.40, and on the wing it is 1.91. Every closure returns between 0.90 and 1.11 in all of these classes. In the outer layer, then, all four are the same curve, a flat line near one, and none of them has any term that responds to the pressure gradient. That single observation accounts for the wing score of -0.393 and the adverse-pressure-gradient boundary-layer score of 0.077: they are the same failure, not two.

Figure 2 is the result we did not expect. Across the ten flat-plate stations the outer-layer ratio is constant at 1.40 while β varies by a factor of five. The fitted slope is zero and the R^2 of that fit is zero. On the wing the ratio does increase with β, by about 0.12 per unit β, but from a higher base of 1.68.

Table 3: Mean P/ε over the outer layer, $0.2 \leq \eta \leq 0.8$.

Flow class	n	Measured	M16	PySR	SEMBAN	SURGA
Channel	13	**0.87**	0.988	0.974	1.090	0.906
Pipe	4	**0.81**	0.983	0.964	1.047	0.923
Couette	3	**0.96**	1.007	0.981	1.023	0.926
ZPG boundary layer	9	**1.10**	0.988	0.977	1.111	0.953
APG boundary layer	10	**1.40**	0.981	0.962	1.037	0.953
Compressible channel	9	**0.90**	1.001	0.971	0.987	0.898
Wing (APG)	4	**1.91**	0.986	0.972	1.088	0.967

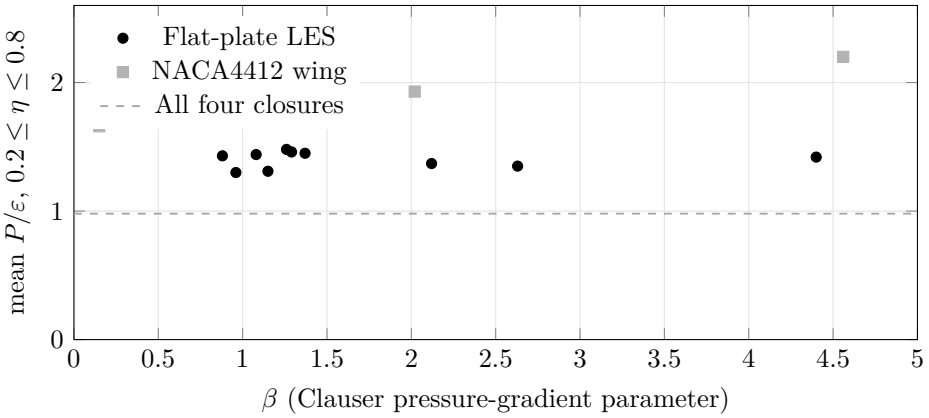


Figure 2: Outer-layer P/ε against pressure-gradient strength. On a flat plate the surplus is constant at about 1.40 while β changes fivefold (fitted slope 0.000, $R^2 = 0.000$). On the wing it grows with β ($1.68 + 0.118\beta$, $R^2 = 0.976$) but starts from a higher value. The closures are the flat dashed line.

This matters for how the closure should be extended. The obvious fifth factor for an adverse-pressure-gradient correction is some function of β. Figure 2 says that on a flat plate such a factor would change nothing, because the target it would try to fit does not depend on β. What is missing is an outer-layer production surplus that switches on when $\mathrm{d}P/\mathrm{d}x > 0$ and is approximately independent of how strong the gradient is. The additional wing surplus is a separate, curvature-related effect.

7 Where SURGA fails

Two failures are worth stating as failures.

Couette. SURGA returns 0.653 on the three Couette profiles while M16 and PySR return 0.887 and 0.897. The cause is visible in the construction. The window factor is squared because $P \propto \tau_R S$ and $S \propto \tau$ in an internal flow. In plane Couette the mean shear is roughly constant across the channel while the total stress falls linearly to zero at the mid-plane, so $S \propto \tau$ is false there. The window over-suppresses the outer half: at $\eta = 0.91$ the measured ratio is 0.85 while the window factor has already fallen to 0.23. The closure is wrong for a reason that can be stated in one line, and the fix would be to replace $S \propto \tau$ with the actual mean shear where it is known.

Adverse pressure gradient. As above. SURGA returns 0.077 on the flat-plate stations, which is a poor score, though better than the negative values of the other closures. The outer surplus of 1.4 is not represented in any of the four factors. The core factor decays with η and the window factor follows τ, and neither of them produces a shoulder.

8 Conclusions

A local closure written as a function of y^+ alone is accurate in the region it was fitted on and wrong over most of the profile. On 52 profiles from nine databases, two well-established y^+-only expressions return negative overall R^2, and they do so for a structural reason that no recalibration can remove: they cannot distinguish two points with the same y^+ and different η. Adding the total stress, the outer scale and the friction Reynolds number to a four-constant closure fixes the bulk of the problem, and does so without being fitted to three of the seven flow classes tested. The remaining failures are specific: plane Couette, where the stress-shear premise fails, and adverse-pressure-gradient flow, where the missing physics is a roughly constant outer-layer surplus rather than anything that depends on the strength of the gradient.

Data availability

All profiles are from public databases cited below and were downloaded from the primary provider. The scoring scripts, the extracted profile data and the table and figure generation code accompany this manuscript.

Acknowledgements

The author thanks the groups listed below for making their simulation data public.

References

[1] R. D. Moser, J. Kim and T. N. Mansour, Direct numerical simulation of turbulent channel flow up to $Re_\tau = 590$, *Phys. Fluids* **11** (1999) 943.

[2] P. Schlatter and R. Örlü, Assessment of direct numerical simulation data of turbulent boundary layers, *J. Fluid Mech.* **659** (2010) 116.

[3] G. K. El Khoury, P. Schlatter, A. Noorani, P. F. Fischer, G. Brethouwer and A. V. Johansson, Direct numerical simulation of turbulent pipe flow at moderately high Reynolds numbers, *Flow Turbul. Combust.* **91** (2013) 475.

[4] M. Bernardini, S. Pirozzoli and P. Orlandi, Velocity statistics in turbulent channel flow up to $Re_\tau = 4000$, *J. Fluid Mech.* **742** (2014) 171.

[5] G. A. Gerolymos and I. Vallet, Pressure, density, temperature and entropy fluctuations in compressible turbulent plane channel flow, *J. Fluid Mech.* **757** (2014) 701.

[6] M. Lee and R. D. Moser, Direct numerical simulation of turbulent channel flow up to $Re_\tau = 5200$, *J. Fluid Mech.* **774** (2015) 395.

[7] A. Bobke, R. Vinuesa, R. Örlü and P. Schlatter, History effects and near-equilibrium in adverse-pressure-gradient turbulent boundary layers, *J. Fluid Mech.* **820** (2017) 667.

[8] M. Lee and R. D. Moser, Extreme-scale motions in turbulent plane Couette flows, *J. Fluid Mech.* **842** (2018) 128.

[9] R. Vinuesa, NACA4412 wing LES database, KTH Mechanics (WingData).

[10] Jiang and Rosić, Combining symbolic regression and turbulence physics to develop a compact inner-layer production–dissipation closure, *Phys. Fluids* **38** (2026) 095140.